



Antony Meckhael

Software Engineer

Software Engineer with experience in backend development, data engineering, and applied machine learning. Proven ability to build scalable backend services, end-to-end data pipelines, and intelligent systems. Co-author of published research in Industry 4.0 digital twins, with hands-on experience in full-stack development, LLM-powered applications, real-time analytics, and mobile platforms.

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EDUCATION

Computer Engineering

GUC (German University in Cairo)

09/2021 - 05/2026

1.03 GPA (Excellent according to German scale)

- **Bachelor thesis (A Graded): "Semantic-Driven Runtime Execution of Containerized Assets in Industry 4.0 Digital Twins"** —Developed at Universität Stuttgart, integrating BaSyx Java SDK with Spring Boot for conditional Docker container execution based on AAS metadata.
- Published as "[A Container-based Approach For Proactive Asset Administration Shell Digital Twins](#)" in collaboration with Universität Stuttgart professors, accepted and discussed at **EDTconf 2025**.

WORK EXPERIENCE

App Developer

Mentalee

05/2025 - 07/2025

AI-based mental wellness platform

Stuttgart, Germany

- Extended a Kotlin Multiplatform application with a focus on iOS integration, including architecture setup (Xcode, Swift bridges) and Unity embedding in the native iOS environment.
- Integrated TensorFlow models into the iOS app for on-device AI features, ensuring performance and reliability.
- Managed source control with GitLab and documented development results and code architecture.
- Collaborated in an agile feature team, contributing to the project's success through problem-solving, feedback, and team support.

FullStack Developer Intern

National Bank of Egypt

07/2024 - 08/2024

- Spearheaded the development and maintenance of a robust project using SQL Server Management Studio (SSMS), ASP.NET, Entity Framework, and Xamarin.Forms, reducing system downtime by 20%.
- Partnered with cross-functional teams to deliver scalable solutions, ensuring a 95% client satisfaction rate.
- Conducted comprehensive debugging and optimization, enhancing application performance by 30%.
- Gained valuable insights from experienced mentors, strengthening both technical and communication skills.

Software Developer

Freelance

01/2024 - 04/2024

- Designed and developed powerful Python-based teaching center software that tracked students' attendance, grades, and payments, automating record-keeping tasks and improving data accuracy for more than 200 students.
- MongoDB is used to keep track of student's records.
- Implemented automated WhatsApp notifications for parents with attendance and grade updates.

SKILLS

- Programming Languages: Java, Python, Kotlin, Swift, C#, JavaScript.
- Backend, Data & Graphs: Spring Boot, RESTful APIs, Apache Spark (PySpark), Kafka, Airflow, PostgreSQL, MongoDB, SQL, Neo4j
- AI, ML & DevOps: Scikit-learn, TensorFlow, LightGBM, NLP, LLMs, SHAP, LIME, Docker, Kubernetes, Git
- Mobile, Foundations & Methods: Kotlin Multiplatform, Swift (iOS), Flutter, React Native | OOP, Data Structures, Algorithms | Agile, Scrum

PERSONAL PROJECTS

Real-Time Stock Analytics Platform (Data Engineering)

- Designed and implemented a real-time data pipeline using Apache Kafka and Spark (PySpark) to process high-velocity trading data.
- Orchestrated distributed ETL workflows with Apache Airflow and deployed services using Docker Compose.
- Built a PostgreSQL data warehouse and a Streamlit-based natural language-to-SQL analytics interface.
- Ensured reproducible, scalable execution through containerized architecture.

Intelligent Hotel Analytics & Graph-RAG System

- Built an end-to-end machine learning pipeline on real hotel, user, and review data, including cleaning, feature engineering, class imbalance handling (SMOTE), and reproducible inference artifacts.
- Detected and resolved severe target leakage using permutation importance and SHAP, validating model behavior with global and local explainability (SHAP, LIME).
- Trained and compared Logistic Regression, Random Forest, LightGBM, and FFNNs, tuning hyperparameters to improve generalization and macro-F1 performance.
- Designed a Neo4j Knowledge Graph and integrated hybrid retrieval (Cypher + embeddings) to ground LLM responses in structured data (Graph-RAG).

Scrum master for Comprehensive Tourism Management Platform (Ranked 1st in Course Project Evaluation)

- Led a team of 10 as Scrum Master, driving Agile practices to ensure on-time delivery of the platform built using the MERN stack (MongoDB, Express.js, React, Node.js).
- Developed a multi-functional web application allowing tourists to book flights, hotels, activities, and tickets, while enabling sellers, tour guides, and advertisers to access tailored functionalities.
- Designed an admin dashboard for efficient content management, user access control, and advertisement management, ensuring smooth operations.

Student management system

- Designed a database-driven student portal using SQL and ASP.NET, improving user engagement for students and teaching assistants.
- The project is a comprehensive database solution that serves both teaching assistants and students. It integrates dual functionality to enhance user engagement and streamline academic processes.

LANGUAGES

Arabic

English

German